

Diagnosis Before Treatment?

Investigating Intermediate-Variable Reasoning in LLMs

Zhishan Yuan¹, Tiago Almeida¹, Shudong Hao¹, Zining Zhu¹,
Nickolas Elias Dawlabani², Ghazia Pervaiz², Yue Ning¹

¹Stevens Institute of Technology, ²Hackensack Meridian School of Medicine

Correspondence: yue.ning@stevens.edu

Abstract

Many expert-domain decisions involve intermediate variables, yet standard LLM evaluation often measures only final-answer accuracy. We present a controlled study of intermediate-variable dependence in medical treatment selection, using diagnosis as an explicit variable between patient phenotype and treatment. We construct a six-option multiple-choice benchmark from PrimeKG-derived phenotype–disease–drug chains and evaluate six LLMs using behavioral diagnosis interventions, held-out QLoRA path supervision, external validation, and auxiliary layer-wise localization analysis. Providing the gold diagnosis improves average treatment accuracy from 49.9% to 84.9%, while incorrect predicted diagnoses reduce accuracy from 85.5% to 23.1%. A shuffled-diagnosis control yields 18.2% accuracy, suggesting that the effect depends on diagnosis correctness rather than merely adding disease-like text. Held-out QLoRA experiments show that path supervision improves treatment selection, with diagnosis-first full-chain and multitask objectives yielding the strongest gains. Center-of-Gravity localization further gives a modest diagnosis-first signal, which we interpret as a localization pattern rather than evidence of faithful clinical reasoning. Overall, in this controlled setting, final-answer accuracy can obscure failures in intermediate-variable dependence, while structured supervision can partially improve diagnosis-to-treatment behavior.

1 Introduction

Many NLP tasks are evaluated as direct input–output mappings, but many expert-domain decisions depend on intermediate variables. A legal judgment may require identifying the relevant issue before deciding a case; a software repair may require localizing the root cause before generating a patch; and a medical treatment decision may require inferring a diagnosis before choosing a treatment. For LLMs, final-answer accuracy alone does

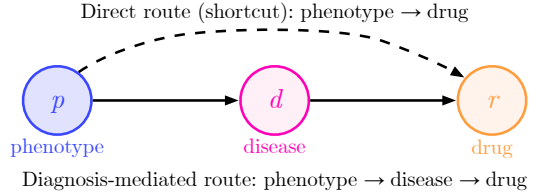


Figure 1: A model may reach the same downstream treatment through different routes: a direct phenotype-to-drug shortcut or a diagnosis-mediated path. We use diagnosis as a domain-aligned intermediate variable for behavioral intervention, path supervision, and auxiliary localization analysis.

not reveal whether the model is sensitive to such domain-aligned variables or instead relies on shortcut associations.

We study this question through medical treatment selection as a controlled testbed for intermediate-variable reasoning. Clinical reasoning is commonly described as a process in which patient observations support candidate diagnoses that guide treatment decisions (Kassirer and Gorry, 1978; Kassirer, 1989; Croskerry, 2002; Scott, 2009). This structure makes diagnosis a natural intermediate variable: given a phenotype, a model may directly associate the phenotype with a drug, or it may infer a disease and then select a treatment determined by that disease as shown in Figure 1. These routes may produce the same final answer, but they imply different error patterns, intervention targets, and localization hypotheses. *Our goal is not to build a clinical decision-support system.* Instead, we use diagnosis-to-treatment selection to study a broader NLP question: when a task has an explicit intermediate variable, can we test whether LLM behavior depends on that variable, whether supervision can improve that dependence, and whether internal localization patterns are compatible with the same staged structure?

Using PrimeKG (Chandak et al., 2023), we con-

struct phenotype–disease–drug chains:

$$p \xrightarrow{\text{disease_phenotype_positive}} d \xrightarrow{\text{indication}} r,$$

where p is a phenotype, d is the hidden diagnosis, and r is a treatment drug. Each chain becomes a six-option treatment selection question in which the disease is known to us but hidden from the direct prompt. This design lets us intervene on diagnosis, supervise different stages of the path, validate the adapter trend on an external source, and test whether patching effects show a compatible layer-wise pattern. We ask three questions:

1. **Behavioral dependence:** Does intervening on the intermediate diagnosis change downstream drug selection? (Section 4.1)
2. **Path supervision:** Does fine-tuning with structured diagnosis-to-treatment supervision improve treatment selection? (Section 4.2)
3. **Auxiliary localization:** Do phenotype-to-disease and disease-to-drug patching effects show a relative layer-wise pattern compatible with the diagnosis-to-treatment structure? (Section 4.3)

To answer these questions, we evaluate six open-weight 7B–8B LLMs with three complementary analyses: behavioral diagnosis interventions, held-out QLoRA path supervision, and contrastive activation patching (Meng et al., 2022; Wang et al., 2023; Conmy et al., 2023), summarized with Center-of-Gravity (CoG) localization (Tenney et al., 2019). We further validate the adapter trend on an independently constructed HPO–CTD benchmark (Köhler et al., 2021; Davis et al., 2025). Diagnosis is strongly consequential in this controlled benchmark: gold diagnosis improves treatment accuracy from 49.9% to 84.9%, while wrong predicted diagnoses reduce accuracy from 85.5% to 23.1%. A shuffled-diagnosis control yields 18.2% accuracy, suggesting the effect depends on diagnosis correctness rather than merely adding disease-like text. QLoRA path supervision improves held-out treatment selection, and the HPO–CTD benchmark shows the same qualitative trend with smaller gains. Contrastive CoG provides weaker but compatible auxiliary evidence: across 590 filtered examples, the average treatment-minus-diagnosis CoG gap is positive (+0.57 layers), and 58.4% of examples show diagnosis-first localization. We summarize our contributions as follows:

- We introduce a controlled phenotype–disease–drug benchmark for studying diagnosis as an explicit intermediate variable in LLM treatment-selection behavior. The benchmark hides the disease during direct treatment prediction but keeps it available for intervention, supervision, and analysis (Section 3).
- We show through diagnosis interventions that the intermediate disease is behaviorally consequential. Gold diagnoses substantially improve treatment selection, wrong predicted diagnoses sharply reduce downstream accuracy, and a shuffled-diagnosis control shows that the effect depends on diagnosis correctness rather than merely adding disease-like text (Section 5.1).
- We show that held-out QLoRA path supervision improves treatment accuracy on chains excluded from adapter training. In the primary data-matched comparison, diagnosis-first full-chain supervision provides an incremental gain over treatment-only supervision, suggesting benefit beyond disease-to-drug training alone. We further validate the same qualitative trend on an independently constructed HPO–CTD benchmark (Sections 4.2, 5.2, and 5.3).
- We use contrastive activation patching and CoG as auxiliary triangulation of relative localization between phenotype-to-disease and disease-to-drug effects. We treat this as a supporting localization signal rather than proof of faithful clinical reasoning (Sections 4.3 and 5.4).

2 Related Work

Intermediate reasoning and faithful evaluation. Chain-of-thought prompting (Wei et al., 2022) and scratchpad methods (Nye et al., 2021) encourage models to produce intermediate reasoning steps before final answers. However, generated rationales are not always faithful to model computation or downstream decision processes (Jacovi and Goldberg, 2020; Turpin et al., 2023). This creates a gap between explanation quality and whether a model actually depends on a domain-relevant intermediate variable. Our work addresses this gap by intervening on an explicit intermediate variable—diagnosis—and measuring its effect on downstream treatment selection.

Intermediate-variable supervision and interventions. A growing line of work argues that final-answer evaluation can miss important structure in

reasoning behavior, and that supervising or intervening on intermediate steps can change both performance and what an evaluation measures (Uesato et al., 2022; Lightman et al., 2024; Paul et al., 2024; Bao et al., 2025). Prior work has studied process supervision, critique-and-refinement, and causal analyses of generated chain-of-thought rationales. Our setting instantiates the $x \rightarrow z \rightarrow y$ structure in medical treatment selection, where phenotype p supports diagnosis d , which in turn determines treatment r . Unlike methods that supervise or analyze intermediate text generated by the model, we treat z as an externally annotated biomedical entity that can be hidden, supplied as an oracle, replaced with an incorrect prediction, supervised through path-specific QLoRA adapters, and localized layer-wise via activation patching. This grounding lets us measure behavioral dependence on the intermediate variable rather than on the surface form of a generated rationale.

Adapter and localization interventions. Parameter-efficient fine-tuning methods such as LoRA (Hu et al., 2022) and QLoRA (Dettmers et al., 2023) enable efficient adaptation without full model retraining. We use QLoRA as a controlled supervision-path intervention, comparing diagnosis-only, treatment-only, diagnosis-first full-chain, and multitask objectives under a shared held-out split. Separately, activation patching and causal tracing estimate how internal representations affect outputs by substituting activations across forward passes (Meng et al., 2022; Wang et al., 2023; Conmy et al., 2023). We use contrastive activation patching only as an auxiliary localization analysis, summarizing relative layer-wise effects with CoG rather than claiming a complete mechanistic circuit.

Medical LLMs and biomedical knowledge graphs. Medical LLMs are commonly evaluated on biomedical and clinical QA benchmarks such as MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022), PubMedQA (Jin et al., 2019), MMLU clinical subsets (Hendrycks et al., 2021), and broader medical LLM evaluations (Singhal et al., 2023; Nori et al., 2023; Arora et al., 2025; Zuo et al., 2025). These benchmarks primarily measure final-answer performance. In contrast, biomedical knowledge graphs such as PrimeKG (Chandak et al., 2023) encode typed phenotype–disease and disease–drug relations, making it possible to construct controlled chains to study intermediate

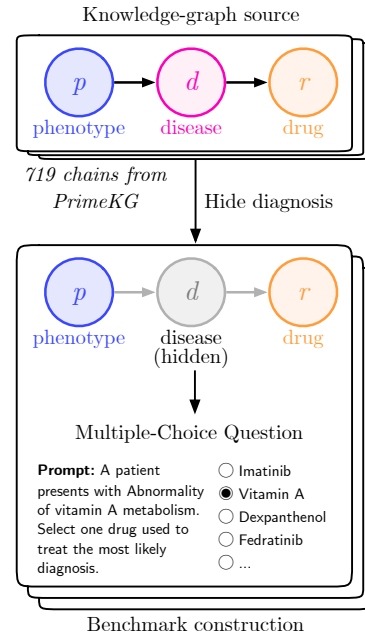


Figure 2: Benchmark construction. We extract phenotype–disease–drug chains from PrimeKG, hide the intermediate disease for downstream prediction, and convert each chain into a six-option question.

variables. We use these chains for model analysis rather than clinical recommendation: diagnosis can be supplied, predicted, supervised, or used for layer-wise localization.

3 Benchmark

We use medical treatment selection as a controlled testbed because its intermediate variable is explicit: phenotype suggests disease, and disease determines treatment. This phenotype–diagnosis–treatment framing is intentionally simplified: it establishes a structured disease–drug relation for model analysis, rather than patient-specific treatment appropriateness or a simulation of real-world clinical reasoning. As shown in Figure 2, we extract phenotype–disease–drug chains from PrimeKG, hide the disease during direct treatment prediction, and convert each chain into a six-option multiple-choice question.

We use PrimeKG because its typed phenotype–disease and disease–drug relations make diagnosis explicit and controllable. The disease entity can be hidden, supplied as an oracle, replaced by a model prediction, used for adapter supervision, or used for layer-wise localization. We extract paths using `disease_phenotype_positive` for phenotype–disease edges and indication for disease–drug edges. We normalize entity names,

remove placeholder or trivially short entities, deduplicate triples, and cap chains per disease to reduce frequency imbalance. The final benchmark contains 719 unique phenotype–disease–drug chains.

Each chain is converted into a six-option multiple-choice question. The model observes the phenotype and selects one drug used to treat the most likely diagnosis. Five distractors are sampled from the global drug pool. We exclude any drug with an indication edge to the same disease, so that no distractor is itself a defensible treatment, and reject candidates sharing a content token with the gold drug, disease, or phenotype, so that the answer cannot be recovered by surface matching. Option order is randomized independently for each question. We score option texts using length-normalized teacher-forced log probability to reduce letter-position and token-length artifacts. A full MCQ example is provided in Appendix A.

To further assess benchmark diversity and potential multiple-choice artifacts, we compute additional construction statistics. The benchmark contains 603 unique phenotypes, 225 diseases, and 193 drugs, with an average of 1.3 drugs per disease and 1.5 diseases per drug. Distractor artifacts are limited: only 0.14% of distractors share any token with the gold drug, and we find no duplicate distractors or cases where the gold drug appears among the distractors. An option-only baseline, which shows only the six drug options without the phenotype, achieves 15.6% accuracy, close to the six-way random baseline of 16.7%. These statistics suggest that superficial answer-option cues are unlikely to explain the main treatment-selection results.

4 Framework

We evaluate the diagnosis-to-treatment chain with three components: behavioral diagnosis interventions, QLoRA path interventions, and auxiliary CoG localization. The three act on different parts of the system. Behavioral interventions manipulate the input while holding parameters fixed, and test whether the annotated intermediate variable changes downstream predictions. QLoRA adapters manipulate parameters while holding the evaluation format fixed, and test whether supervision on different segments of the path steers held-out behavior. CoG localization manipulates internal activations, and checks whether phenotype-to-disease and disease-to-drug patching effects show a relative layer-wise pattern compatible with the same

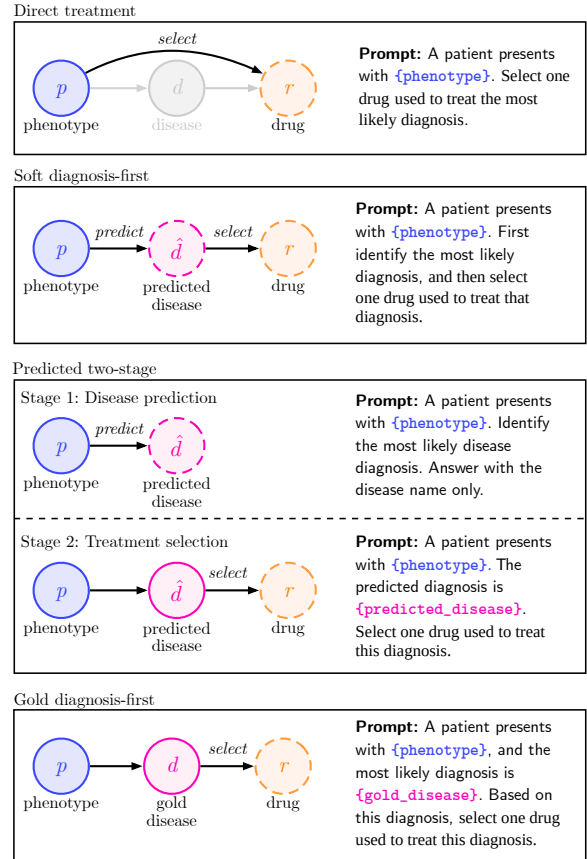


Figure 3: Behavioral diagnosis interventions. The figure shows the four main settings: direct treatment, soft diagnosis-first prompting, predicted two-stage prompting, and gold-diagnosis prompting. We additionally use a shuffled-diagnosis control, not shown in the figure, to test whether the effect depends on diagnosis correctness rather than merely adding disease-like text.

staged structure. We treat the three analyses as triangulation rather than interchangeable evidence.

4.1 Diagnosis-Intervention Evaluation

We design four main diagnosis-intervention settings to test whether the intermediate diagnosis changes downstream treatment selection, together with a shuffled-diagnosis control. As shown in Figure 3, the settings differ in whether diagnosis is absent, encouraged only by instruction, predicted by the model, or provided as an oracle variable.

In the direct treatment setting, the model receives only the phenotype and directly selects a drug from the answer options. This tests phenotype-to-drug behavior without supplying any concrete diagnosis string as an input variable. Although the direct prompt refers to the most likely diagnosis to define the treatment relation, the diagnosis itself is not provided to the model. In the *soft diagnosis-first*

setting, the prompt asks the model to first reason about the most likely diagnosis and then choose a treatment, but no diagnosis is inserted into the input. This setting tests whether instructional phrasing alone can encourage a diagnosis-mediated route.

In the *predicted two-stage* setting, we explicitly decompose the task into two steps. The model first predicts a disease $\hat{d}$ from the phenotype, and this predicted diagnosis is then inserted into a second treatment-selection prompt. This setting tests whether an explicit model-generated intermediate variable can improve treatment selection, and whether errors in diagnosis propagate to treatment. In the *gold diagnosis-first* setting, we replace the predicted diagnosis with the gold disease d and insert it into the treatment prompt as an oracle intervention. This provides an upper-bound test of how much treatment selection can improve when the intermediate diagnosis is correct.

We additionally use a *shuffled-diagnosis* control to separate the effect of diagnosis correctness from the effect of adding disease text. This control keeps the same phenotype, answer options, and gold treatment, but replaces the gold disease with a disease sampled from another example. If gains were due merely to extra disease-like context, the shuffled diagnosis should also help; if correctness matters, it should instead misdirect treatment selection.

Across all settings, we report top-1 accuracy, correct-option probability, and probability margin. Full prompt templates are provided in Appendix E.

4.2 QLoRA Path Interventions

We use QLoRA as a lightweight supervision-path intervention for three reasons. First, it keeps the base model frozen and updates only low-rank adapter parameters, making the intervention more controlled than full fine-tuning. Second, 4-bit NF4 quantization makes adaptation practical for six 7B–8B models under a shared experimental setup. Third, using separate adapters lets us compare supervision paths while holding the base model fixed. Thus, we use QLoRA as an intervention tool rather than as a claim about optimal medical fine-tuning.

For each chain (p, d, r) , we train four objectives: diagnosis-only ($p \rightarrow d$), treatment-only ($d \rightarrow r$), diagnosis-first full-chain ($p \rightarrow d \rightarrow r$), and multitask diagnosis-first supervision. These objectives compare whether gains come from diagnostic inference, disease-to-drug mapping, or their combination. Because the multitask adapter contains multiple formats per chain, we treat it as a

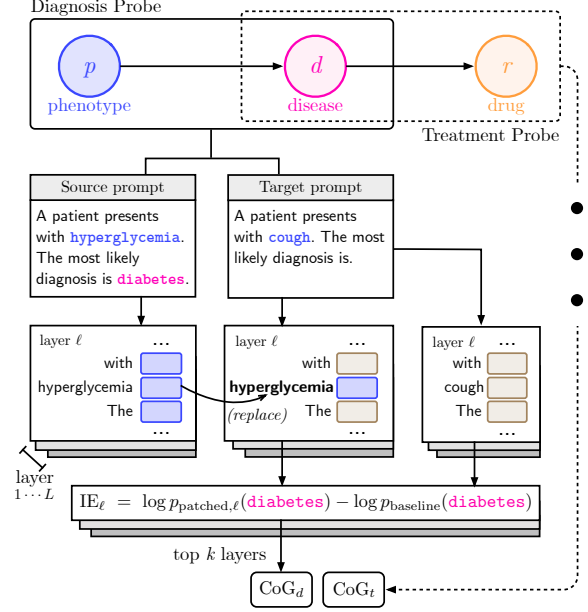


Figure 4: Activation-patching pipeline for CoG localization. The figure illustrates the diagnosis probe: phenotype-span activations from a source prompt are patched into a target prompt at each layer, and the indirect effect on the gold disease probability is summarized by a softmax-weighted CoG. The treatment probe applies the same procedure to disease-span activations and gold-drug probability.

practical mixed-objective intervention; the primary data-matched comparison is treatment-only versus diagnosis-first full-chain.

All adapters use the same chain-level split: 503 train, 71 dev, and 145 held-out test chains. Training uses only train chains, and MCQ evaluation uses only held-out test chains. QLoRA hyperparameters are provided in Appendix B, and prompt templates are provided in Appendix E.

4.3 Auxiliary CoG Localization

We use activation patching as an auxiliary localization analysis for the diagnosis-to-treatment structure of the task. The goal is to test whether the behavioral diagnosis-to-treatment dependence observed above has a compatible internal localization pattern in LLMs.

For each phenotype–disease–drug chain, we construct two cloze probes: a diagnosis probe for phenotype-to-disease localization and a treatment probe for disease-to-drug localization. In each probe, we patch a source-span representation into the corresponding target span one layer at a time and measure the change in teacher-forced log probability of the gold answer as shown in Figure 4.

Diagnosis probe: phenotype to disease. The diagnosis probe estimates where phenotype information most affects disease prediction.

Diagnosis probe
Source prompt: A patient presents with <phenotype>. The most likely diagnosis is <disease>.
Target prompt: A patient presents with <alternative phenotype>. The most likely diagnosis is —

We cache the mean hidden representation of the phenotype span from the source prompt, patch it into the phenotype span of the target prompt at each layer, and measure the change in teacher-forced log probability of the gold disease.

Treatment probe: disease to drug. The treatment probe estimates where disease information most affects drug prediction.

Treatment probe
Source prompt: For <disease>, one standard drug is <drug>.
Target prompt: For <alternative disease>, one standard drug is —

We cache the mean hidden representation of the disease span from the source prompt, patch it into the disease span of the target prompt at each layer, and measure the change in teacher-forced log probability of the gold drug.

For layer l , the indirect effect is

$$\text{IE}_l = \log p_{\text{patched},l}(y) - \log p_{\text{baseline}}(y), \quad (1)$$

where y is the gold disease for the diagnosis probe and the gold drug for the treatment probe. We summarize each IE curve with a softmax-weighted Center of Gravity over the top- k layers, where TopK denotes the k layers with the largest indirect effects:

$$\text{CoG} = \sum_{l \in \text{TopK}} w_l \times l, \quad (2)$$

$$w_l = \frac{\exp(\text{IE}_l/\tau)}{\sum_{j \in \text{TopK}} \exp(\text{IE}_j/\tau)}. \quad (3)$$

We set $k=5$ to average over the strongest patching-effect layers rather than relying on a single argmax layer, which can be sensitive to local noise in the IE curve. We set $\tau=1.0$ as a neutral softmax temperature. Both hyperparameters are fixed across all models and examples.

We define $\text{CoG}_{\text{diagnosis}}$ as the CoG from the diagnosis probe and $\text{CoG}_{\text{treatment}}$ as the CoG from the treatment probe. Our primary ordering score is

$$\Delta \text{CoG}_{t-d} = \text{CoG}_{\text{treatment}} - \text{CoG}_{\text{diagnosis}}. \quad (4)$$

Model	Diag	Dir.	Soft	2S	Gold	G-D
Meditron3	.679	.481	.473	.652	.860	+.379
OpenBioLLM	.565	.428	.314	.548	.800	+.372
Med42	.725	.498	.474	.694	.855	+.357
HuatuoGPT	.691	.529	.523	.668	.868	+.339
Qwen2.5	.686	.529	.483	.669	.840	+.311
Llama3.1	.704	.531	.533	.682	.872	+.341
Average	.675	.499	.467	.652	.849	+.350

Table 1: Gold-diagnosis intervention substantially improves treatment selection for all models on the full 719-chain PrimeKG benchmark. Diag is phenotype-to-disease accuracy; Dir. is direct phenotype-to-drug selection; Soft is diagnosis-first prompting without the gold diagnosis; 2S inserts the model-predicted diagnosis; Gold inserts the gold diagnosis; G-D is Gold minus Direct.

A positive ΔCoG_{t-d} indicates diagnosis-first localization, meaning that treatment-related effects peak later than diagnosis-related effects. We interpret this as an auxiliary localization signal rather than proof of faithful clinical reasoning.

4.4 Models

We evaluate six open-weight 7B–8B-scale models: Qwen2.5-7B-Instruct (Qwen Team, 2025), Llama3.1-8B-Instruct (Llama Team, AI @ Meta, 2024), Llama3-OpenBioLLM-8B (Saama AI Labs, 2024), Meditron3-8B (Sallinen et al., 2025), Med42-8B (Christophe et al., 2024), and HuatuoGPT-o1-8B (Chen et al., 2025). We choose this scale because it supports both QLoRA adaptation and activation patching with hidden-state access under a shared setup. The set includes general instruction models and medical-domain adapted models, allowing us to compare whether domain specialization affects diagnosis-to-treatment behavior.

5 Experimental Results and Analyses

5.1 Behavioral Diagnosis Interventions

We first test whether diagnosis changes downstream treatment behavior. A meaningful intermediate variable should satisfy three conditions in our controlled setting: an oracle value should improve the final prediction, model-generated errors should propagate downstream, and a soft instruction should be weaker than explicitly inserting the intermediate variable.

Diagnosis interventions. Table 1 shows three patterns. First, gold diagnosis improves treatment

Model	$P(\text{Drug} \text{Diag} \checkmark)$	$P(\text{Drug} \text{Diag} \times)$	Gap
Meditron3	0.859	0.216	0.643
OpenBioLLM	0.815	0.201	0.614
Med42	0.856	0.268	0.588
HuatuoGPT	0.865	0.225	0.640
Qwen2.5	0.860	0.252	0.608
Llama3.1	0.875	0.221	0.654
Average	0.855	0.231	0.624

Table 2: Treatment accuracy is strongly conditioned on diagnosis correctness. Correct diagnoses yield high treatment accuracy, whereas wrong diagnoses sharply reduce downstream performance.

accuracy for every model, with an average gain of +35.0 points. Second, predicted two-stage evaluation improves over direct treatment by +15.3 points on average, but remains 19.7 points below gold diagnosis. Third, soft diagnosis-first prompting does not reliably help, decreasing average accuracy from 0.499 to 0.467. Thus, models benefit from accurate intermediate information, but instruction alone is insufficient.

Error propagation. Table 2 shows that treatment accuracy is strongly coupled to diagnosis correctness. When the predicted diagnosis is correct, treatment accuracy averages 0.855; when it is wrong, accuracy drops to 0.231. Together, these results support diagnosis as a behaviorally consequential intermediate variable: oracle diagnosis improves treatment selection, predicted diagnoses partially recover the pathway, and wrong diagnoses sharply degrade downstream accuracy.

Diagnosis-correctness control. To test whether the gold-diagnosis gain is due merely to adding disease-like text, we add a shuffled-diagnosis control. This setting keeps the same phenotype, answer options, and gold treatment, but replaces the gold disease with a disease sampled from another example. Shuffled diagnoses reduce treatment accuracy to 18.2%, below the direct baseline of 49.9%, whereas gold diagnoses increase accuracy to 84.9%. The Gold–Shuffled gap is +66.7 points, and the Shuffled–Direct gap is −31.7 points. This indicates that the intervention effect depends on the correctness of the inserted diagnosis, not merely on the presence of additional disease-like text. We interpret this as evidence of controlled behavioral dependence on the intermediate variable, while not ruling out retrieval of disease–drug associations.

5.2 QLoRA Path Interventions

We next test whether diagnosis-to-treatment path supervision can improve downstream treatment selection beyond prompting alone. Unlike the behavioral intervention results in Table 1, which are reported on the full 719-chain benchmark, this section evaluates all adapters on the 145 held-out adapter test chains. Thus, the direct/base accuracies in the two sections are not expected to match. All adapters are trained on training chains and evaluated on held-out test chains that are excluded from adapter training. This held-out split tests whether adapter gains transfer to unseen phenotype–disease–drug chains rather than reflecting evaluation-set memorization.

Figure 5 shows that diagnosis-first path supervision consistently improves held-out treatment selection. Diagnosis-first full-chain supervision raises average accuracy from 0.37 to 0.56 (+19.4 points). The improvement appears across all six models, suggesting that training the model to follow the phenotype-to-diagnosis-to-treatment path can improve treatment selection on unseen chains.

Multitask diagnosis-first training achieves the highest average accuracy, 0.57 (+20.2 points), but it uses multiple formats per chain. We therefore treat it as a practical mixed-objective variant rather than the primary data-matched comparison.

The primary data-matched comparison is treatment-only versus diagnosis-first full-chain supervision. Treatment-only reaches 0.5264 average accuracy, confirming that disease-to-drug mapping is a strong learnable component of the task. Diagnosis-first full-chain supervision further improves accuracy to 0.5621 and outperforms treatment-only in 5 of the 6 models. A paired-bootstrap analysis over the 145 held-out chains shows positive macro-averaged differences for accuracy (+0.0356, 95% CI [+0.0115, +0.0609]), correct-option probability (+0.0239, 95% CI [+0.0167, +0.0312]), and probability margin (+0.0270, 95% CI [+0.0153, +0.0387]).

These results suggest an incremental benefit from full phenotype-to-diagnosis-to-treatment supervision beyond disease-to-drug training alone, while remaining within a controlled chain-level evaluation. Complete adapter results are provided in Appendix G.

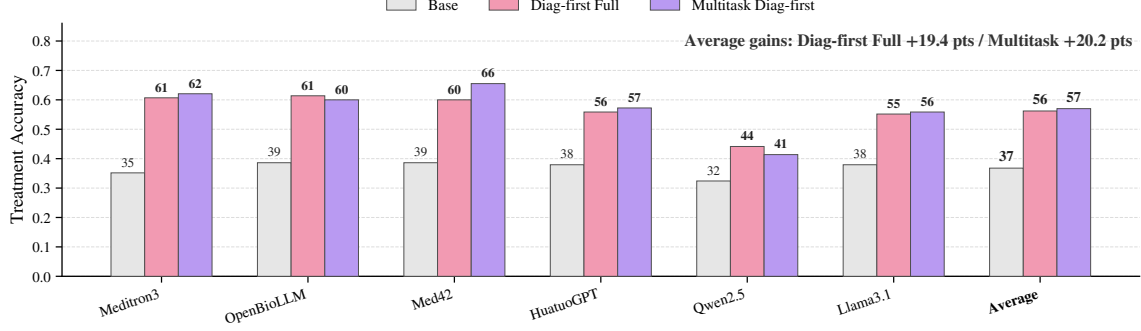


Figure 5: Held-out adapter evaluation on test chains excluded from adapter training. The figure highlights diagnosis-first path supervision, which substantially improves treatment accuracy over base models. The final group reports the average across six models.

5.3 External Validation on HPO-CTD

To test whether the adapter trend is specific to the PrimeKG-derived benchmark, we evaluate the same adapters on an independently constructed HPO-CTD benchmark. This set is built by linking HPO phenotype-disease annotations (Köhler et al., 2021) with CTD therapeutic drug-disease associations (Davis et al., 2025) and converting the resulting phenotype-disease-drug triples into the same six-option treatment-selection format. We use this experiment as a robustness check rather than as clinical validation.

Setting	n	Acc	Δ Acc	Prob	Δ Prob
Base	286	.302	–	.232	–
Diagnosis-only	286	.300	–.002	.230	–.002
Treatment-only	286	.321	+.019	.247	+.015
Diagnosis-first full	286	.336	+.034	.253	+.021
Multitask diag-first	286	.329	+.027	.254	+.022

Table 3: Adapter evaluation on the HPO-CTD benchmark. Results are macro-averaged across six models on 286 test examples. Diagnosis-first full-chain supervision improves average accuracy from 0.302 to 0.336.

Table 3 shows the same trend as the PrimeKG held-out evaluation, although the gains are smaller. Diagnosis-first full-chain supervision improves average accuracy from 0.302 to 0.336, with the same qualitative trend as the PrimeKG held-out evaluation. Multitask diagnosis-first supervision reaches 0.329. We interpret this as external robustness evidence for the supervision-path finding.

5.4 Auxiliary Layer-wise Analysis

Finally, we use contrastive CoG localization as a triangulation analysis rather than a standalone proof of reasoning order. The goal is to ask whether the

behavioral diagnosis-to-treatment dependence observed above has a compatible internal localization pattern under a controlled patching procedure. For each filtered example, we compute ΔCoG_{t-d} as defined in Eq. 4, and report the macro-average across six models.

Metric	Macro Avg.
Filtered examples	590
Raw examples	1988
ΔCoG_{t-d}	+0.57
Diagnosis-first ratio	58.4%

Table 4: Macro-averaged two-probe CoG localization results across six models. ΔCoG_{t-d} is computed as $\text{CoG}_{\text{treatment}} - \text{CoG}_{\text{diagnosis}}$. Diagnosis-first localization means $\text{CoG}_{\text{diagnosis}} < \text{CoG}_{\text{treatment}}$.

Table 4 shows a mild diagnosis-first tendency. After filtering weak or unstable indirect-effect curves, 590 examples are retained from 1988 raw examples. The mean treatment-minus-diagnosis CoG gap is positive, and 58.4% of filtered examples show diagnosis-first localization. These patterns are consistent with the broader diagnosis-to-treatment story, but they are not deterministic, so we interpret CoG ordering as auxiliary localization evidence rather than as a causal explanation of correctness.

6 Discussion

Intermediate variables as evaluation objects.

Our results support treating intermediate variables as first-class objects in LLM evaluation. In direct evaluation, treatment selection is measured only as a final answer. Our benchmark instead keeps diagnosis available as an explicit variable that can be hidden, predicted, supplied, supervised, and analyzed. This reveals behavior that final-answer accu-

racy cannot separate: gold diagnoses substantially improve treatment selection, while incorrect predicted diagnoses sharply reduce downstream performance. Thus, in this testbed, diagnosis is not merely a post-hoc explanation but a behaviorally consequential intermediate variable.

Path supervision is useful but not sufficient.

Held-out QLoRA results show that structured supervision can steer treatment selection on unseen chains. Diagnosis-first full-chain and multitask adapters yield the strongest average gains, while treatment-only controls also improve performance. The additional HPO-CTD results show the same qualitative trend with smaller gains, suggesting that the supervision-path effect is not limited to the PrimeKG split. This pattern suggests that both phenotype-to-diagnosis information and disease-to-drug mapping contribute to downstream behavior. However, these gains should not be interpreted as proof of faithful clinical reasoning: adapters improve the model’s behavior in controlled knowledge-graph-derived tasks, but they do not establish that the model follows the same reasoning process as a clinician.

Localization is supporting evidence. The two-probe contrastive CoG analysis provides a compatible but modest internal signal: diagnosis-first localization appears in a small majority of filtered examples. Because CoG is a layer-wise summary and the effect is not deterministic, we use it as auxiliary triangulation rather than primary evidence. The main evidence for intermediate-variable dependence comes from behavioral interventions, held-out path supervision, and external validation.

Beyond medicine. Although our experiments focus on diagnosis-to-treatment selection, the same intervention logic applies to structured tasks of the form $x \rightarrow z \rightarrow y$, where z is an explicit intermediate variable. In legal reasoning, z may be the relevant issue or statute; in software debugging, it may be the root cause; and in scientific QA, it may be the relevant mechanism or evidence relation. In these settings, final-answer accuracy can hide whether the model is sensitive to the intermediate variable. The key requirement is that z be independently annotatable and intervention-ready.

More broadly, our study suggests a path for future NLP evaluation: rather than treating expert tasks only as input-output benchmarks, we can design testbeds around intermediate variables.

Such benchmarks make it possible to measure intermediate-variable correctness, test interventions, compare supervision paths, and examine compatible localization signals. This shifts evaluation from asking only whether a model produces the correct final answer toward asking whether its behavior is sensitive to the intermediate variables that define a structured decision problem.

7 Conclusion

We present a controlled testbed for intermediate-variable reasoning in LLM treatment selection, using diagnosis as an explicit intermediate variable between patient phenotype and treatment choice.

Across six models, diagnosis is behaviorally consequential: providing the gold diagnosis substantially improves treatment accuracy, while wrong predicted diagnoses sharply reduce downstream performance. Held-out QLoRA path supervision further improves treatment selection, with gains reflecting both diagnostic inference and disease-to-drug mapping. An independently constructed HPO-CTD benchmark shows the same qualitative supervision-path trend with smaller gains. As an auxiliary internal analysis, contrastive CoG provides a modest localization signal, suggesting that phenotype-to-disease and disease-to-drug patching effects show a compatible relative ordering, though not a complete mechanistic circuit.

Overall, our results show that final-answer accuracy alone can obscure whether a model is sensitive to domain-aligned intermediate variables. For structured expert tasks, evaluation should test not only whether a model reaches the right answer, but also whether the intermediate variable is correct, behaviorally consequential, steerable through supervision, and compatible with internal evidence.

Acknowledgement

This research was funded in part by the US National Science Foundation under grants IIS-2047843 and IIS-2437621. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the NSF.

Limitations

Our benchmark is intentionally controlled: each example contains a single phenotype, one hidden disease, and one documented treatment edge from

PrimeKG. This design isolates the phenotype-to-diagnosis-to-treatment chain, but it does not model full clinical practice, where decisions may depend on multiple symptoms, tests, comorbidities, contraindications, and differential diagnoses. The external benchmarks provide robustness checks, but they are also constructed from structured biomedical resources and do not replace expert clinical cases. Held-out adapter evaluation removes direct chain overlap, but the split is chain-level rather than entity-level; diseases and drugs may appear across train and test chains. Therefore, our held-out results should be interpreted as chain-level generalization, not disease-level or drug-level generalization. Although the shuffled-diagnosis control shows that diagnosis correctness matters, it does not rule out retrieval of memorized disease–drug associations. Multitask adapters contain more training instances than single-task adapters, so their gains should be interpreted as mixed-objective results. Finally, CoG is only a relative layer-wise localization summary based on two contrastive probes. The benchmark should therefore be read as a controlled model-analysis setting, not a clinical recommendation system.

Ethics and Risks. Our benchmark is constructed from curated biomedical knowledge graphs and contains no personal health information, patient records, or identifiable data. The intermediate-variable analysis is designed for studying LLM behavior, not for clinical decision support, and we want to be explicit about the risks of misinterpreting our results.

Misuse for clinical decisions. The phenotype, disease, and drug chains in our benchmark are simplified, single-edge instances drawn from knowledge-graph relations. They do not represent differential diagnosis, comorbidities, contraindications, dosing, or patient-specific factors. The treatment-accuracy numbers reported here, including the gold-diagnosis upper bound, should not be read as evidence that any model is safe or suitable for clinical use. We strongly discourage deployment of the released adapters in any setting that influences real patient care.

Coverage and representational bias. PrimeKG, HPO, and CTD integrate predominantly Western, English-language biomedical literature and ontologies. Diseases, phenotypes, and drug indications that are under-represented in these sources, including conditions more prevalent in low-resource settings and traditional or non-Western therapeutic

tics, are correspondingly under-represented in our chains. Behavioral and supervision results therefore inherit these coverage biases, and improvements observed here may not transfer to under-represented conditions or populations. Medical LLMs known to encode such biases (Omiye et al., 2023; Singhal et al., 2023) are part of our evaluation set, and our findings should not be used to argue otherwise.

Dual use of path supervision. The QLoRA path-supervision recipe we study could in principle be applied to push other models toward diagnosis-first behavior on clinical-style inputs. Because our adapters improve held-out accuracy on a controlled benchmark without establishing faithful clinical reasoning, third parties could overstate adapter-trained models as clinically reliable. We release adapters and code for research transparency only, with explicit research-only terms and a documented intended-use statement.

Environmental cost. Training four adapter objectives across six 7B to 8B base models, together with layer-wise activation patching across all transformer layers, has a non-trivial compute footprint. We mitigate this by using 4-bit QLoRA rather than full fine-tuning and by sharing a single chain-level split across all adapters and models. Training hyperparameters are reported in Appendix B.

Stakeholders. The intended audience for this work is NLP and ML researchers studying intermediate-variable reasoning and faithful evaluation, not clinicians, patients, or healthcare administrators. We do not recommend that any clinical or policy decisions be made on the basis of the results reported here.

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Appendix

A MCQ Example

Field	Example
Chain	Focal aware seizure → herpes simplex encephalitis → Acyclovir
Prompt	A patient presents with Focal aware seizure. Select one drug used to treat the most likely diagnosis.
Options	A. Ursodeoxycholic acid B. Acyclovir C. Acenocoumarol D. Fusidic acid E. Sodium stibogluconate F. Testosterone undecanoate
Gold answer	B. Acyclovir
Hidden diagnosis	herpes simplex encephalitis

Table 5: A complete MCQ example derived from a phenotype–disease–drug chain. The model observes only the phenotype and drug options; the diagnosis is hidden during direct treatment prediction and used only for intervention and analysis.

B QLoRA Configuration

Setting	Value
Quantization	4-bit NF4
Adapter method	QLoRA
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
Max sequence length	512
Per-device batch size	1
Gradient accumulation	16 steps
Learning rate	2×10^{-4}
Warmup ratio	0.03
LR scheduler	Cosine
Precision	fp16
Training epochs	6

Table 6: QLoRA training configuration used for all adapter objectives.

C Fine-tuning Examples

The adapters are trained on instruction–response examples. Prompt tokens are masked, and loss is applied only to response tokens.

Treatment-only Control

Prompt:
You are a medical reasoning assistant. A disease diagnosis is given below. Your task is to provide one drug used to treat this

disease.
Disease: familial Mediterranean fever
Return only the treatment in exactly this format:
Drug: <drug name>
Response:
Drug: Canakinumab

Diagnosis-first Full-chain

Prompt:
You are a medical reasoning assistant.
A patient presentation is given below. First identify the most likely disease diagnosis. Then provide one drug used to treat that disease.
Patient presentation: Macroorchidism
Return the answer in exactly this format:
Disease: <disease name>
Drug: <drug name>
Response:
Disease: familial male-limited precocious puberty
Drug: Histrelin

Diagnosis-only

Prompt:
You are a medical reasoning assistant.
A patient presentation is given below. Your task is to identify the most likely disease diagnosis.
Patient presentation: Abnormality of the femoral metaphysis
Return only the diagnosis in exactly this format:
Disease: <disease name>
Response:
Disease: bone osteosarcoma

D Two-stage Evaluation

Predicted two-stage evaluation decomposes treatment selection into an explicit diagnosis step followed by a treatment step. The intermediate disease is first predicted by the same model from the phenotype and then inserted into a second-stage treatment prompt. This separates two questions: whether explicit diagnostic routing helps over direct treatment, and how much gap remains due to phenotype-to-disease prediction mistakes.

E Prompt Templates

The following templates illustrate the prompting and continuation-scoring formats used in our experiments. For option-text scoring, each candidate drug option is evaluated as a continuation under the same prompt.

Direct Treatment

A patient presents with {phenotype}. Select one drug used to treat the most likely diagnosis. Answer with only the drug name.

{options}

Answer:

Soft Diagnosis-first

A patient presents with {phenotype}. Use a diagnosis-first clinical reasoning strategy. First identify the most likely diagnosis, then select one drug used to treat that diagnosis. Answer with only the drug name.

{options}

Answer:

Predicted Two-stage: Diagnosis Prediction

A patient presents with {phenotype}. Identify the most likely disease diagnosis. Answer with only the disease name.

Diagnosis:

Predicted Two-stage: Treatment Selection

A patient presents with {phenotype}. The predicted diagnosis is {predicted_disease}. Based on this diagnosis, select one drug used to treat this diagnosis. Answer with only the drug name.

{options}

Answer:

Gold Diagnosis-first

A patient presents with {phenotype}. The most likely diagnosis is {gold_disease}. Based on this diagnosis, select one drug used to treat this diagnosis. Answer with only the drug name.

{options}

Answer:

Shuffled Diagnosis Control

A patient presents with {phenotype}. The most likely diagnosis is {shuffled_disease}. Based on this diagnosis, select one drug used to treat this diagnosis. Answer with only the drug name.

{options}

Answer:

F Additional OptimusKG Validation

We additionally evaluate the adapters on an OptimusKG (Vittor et al., 2026)-derived benchmark. We use this dataset as a supplementary robustness

check and do not include it in the main conclusions.

Setting	<i>n</i>	Acc	Δ Acc	Prob	Δ Prob
Base	301	.330	–	.261	–
Diagnosis-only	301	.330	+0.000	.257	–.004
Treatment-only	301	.342	+0.012	.257	–.004
Diagnosis-first full	301	.348	+0.018	.262	+0.001
Multitask diag-first	301	.353	+0.023	.262	+0.001

Table 7: Additional validation on the OptimusKG-derived benchmark. Results are macro-averaged across six models on 301 test examples. This dataset is used as a supplementary robustness check and is not part of the main conclusions.

G Full Held-out Adapter Results

Table 8 reports the complete held-out adapter evaluation for every base model and adapter objective. All rows use the same 145 held-out test chains, which are excluded from adapter training. Δ Acc and Δ Prob are computed relative to the corresponding base model.

H License

All scientific artifacts used in this work are credited in Sections 1, 3, 4.4. We use these artifacts for non-commercial academic research, which is consistent with their intended use.

- **Models.** Qwen2.5-7B-Instruct is released under the Apache 2.0 license. Llama-3.1-8B-Instruct and Meditron3-8B are released under the Llama 3.1 Community License Agreement. Llama3-OpenBioLLM-8B and Llama3-Med42-8B are derivatives of Llama-3-8B and Llama-3-8B-Instruct, respectively, and are released under the Llama 3 Community License Agreement. HuatuoGPT-o1-8B is under the Apache 2.0 license and is subject to the Llama 3.1 Community License Agreement. We use all models in accordance with their license terms and acceptable-use policies, for non-commercial academic research consistent with their stated intended use.
- **Datasets.** PrimeKG (Chandak et al., 2023) and OptimusKG (Vittor et al., 2026) are distributed on Harvard Dataverse; their construction code is released under MIT, while the integrated knowledge graph data inherits the licenses of its upstream sources. The Human Phenotype Ontology (Köhler et al., 2021) is released under CC BY 4.0. The Comparative Toxicogenomics Database (Davis et al., 2025) is freely available

for academic research under CTD's terms of use, with required citation. We use all datasets in accordance with these terms.

- Released benchmark. We will release the benchmark construction code under the MIT license. The derived phenotype, disease, and drug chains will be released under CC BY 4.0 to the extent permitted by upstream licenses, with explicit attribution to PrimeKG, HPO, and CTD, and with a note that downstream users must additionally honor any restrictions imposed by these upstream sources. The benchmark is intended for research on intermediate-variable reasoning in LLMs and is not intended for clinical use.

Model	Version	n	Acc	Δ Acc	Correct Prob	Δ Prob
Meditron3	Base	145	0.3517	–	0.2541	–
Meditron3	Diagnosis-only	145	0.3517	+0.0000	0.2356	–0.0185
Meditron3	Diagnosis-first full	145	0.6069	+0.2552	0.3739	+0.1198
Meditron3	Treatment-only	145	0.5517	+0.2000	0.3326	+0.0785
Meditron3	Multitask diag-first	145	0.6207	+0.2690	0.3709	+0.1168
OpenBioLLM	Base	145	0.3862	–	0.2990	–
OpenBioLLM	Diagnosis-only	145	0.4345	+0.0483	0.3136	+0.0146
OpenBioLLM	Diagnosis-first full	145	0.6138	+0.2276	0.4004	+0.1014
OpenBioLLM	Treatment-only	145	0.5241	+0.1379	0.3699	+0.0709
OpenBioLLM	Multitask diag-first	145	0.6000	+0.2138	0.3936	+0.0946
Med42	Base	145	0.3862	–	0.2914	–
Med42	Diagnosis-only	145	0.4138	+0.0276	0.2951	+0.0037
Med42	Diagnosis-first full	145	0.6000	+0.2138	0.3715	+0.0801
Med42	Treatment-only	145	0.5862	+0.2000	0.3482	+0.0568
Med42	Multitask diag-first	145	0.6552	+0.2690	0.3900	+0.0986
HuatuoGPT	Base	145	0.3793	–	0.2730	–
HuatuoGPT	Diagnosis-only	145	0.4069	+0.0276	0.2732	+0.0002
HuatuoGPT	Diagnosis-first full	145	0.5586	+0.1793	0.3565	+0.0835
HuatuoGPT	Treatment-only	145	0.5448	+0.1655	0.3433	+0.0703
HuatuoGPT	Multitask diag-first	145	0.5724	+0.1931	0.3531	+0.0801
Qwen2.5	Base	145	0.3241	–	0.2310	–
Qwen2.5	Diagnosis-only	145	0.3310	+0.0069	0.2435	+0.0125
Qwen2.5	Diagnosis-first full	145	0.4414	+0.1173	0.2736	+0.0426
Qwen2.5	Treatment-only	145	0.3655	+0.0414	0.2405	+0.0095
Qwen2.5	Multitask diag-first	145	0.4138	+0.0897	0.2638	+0.0328
Llama3.1	Base	145	0.3793	–	0.2648	–
Llama3.1	Diagnosis-only	145	0.4207	+0.0414	0.2686	+0.0038
Llama3.1	Diagnosis-first full	145	0.5517	+0.1724	0.3395	+0.0747
Llama3.1	Treatment-only	145	0.5862	+0.2069	0.3378	+0.0730
Llama3.1	Multitask diag-first	145	0.5586	+0.1793	0.3375	+0.0727
Average	Base	145	0.3678	–	0.2689	–
Average	Diagnosis-only	145	0.3931	+0.0253	0.2716	+0.0027
Average	Diagnosis-first full	145	0.5621	+0.1943	0.3526	+0.0837
Average	Treatment-only	145	0.5264	+0.1586	0.3287	+0.0598
Average	Multitask diag-first	145	0.5701	+0.2023	0.3515	+0.0826

Table 8: Complete held-out adapter evaluation on the 145-chain PrimeKG test split. All rows use the same held-out adapter test chains, which are excluded from adapter training. Δ Acc and Δ Prob are computed relative to the corresponding base model.